

[Title Task 1 - Email Spam Classification Report]

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1.Introduction & Objective :

1.1 Objective: The objective of this project is to construct an automated Machine Learning classifier that categorizes text messages and emails as either **Spam** or **Ham** (Legitimate). Using Natural Language Processing (NLP) techniques, raw text data is transformed into numerical features for efficient model training and accurate prediction.

1.2 Technologies Used: Python, Scikit-learn, Pandas, TF-IDF Vectorizer, Multinomial Naive Bayes.

2.Methodology & Implementation:

2.1 Dataset: Kaggle SMS Spam Collection dataset.

2.2 Label Mapping: Target string categories (ham/spam) were encoded into binary integers (0 for Ham, 1 for Spam).

2.3 Feature Extraction: Used TfidfVectorizer (Term Frequency-Inverse Document Frequency) with English stop-word removal to extract key textual features into vector representations.

2.4 Preprocessing: Mapped textual targets (ham/spam) to numerical binary labels (0/1) and extracted TF-IDF features with English stop-word filtering.

2.5 Model: Trained a Multinomial Naive Bayes classifier on an 80/20 train-test split.

3.Code Implementation:

```
from google.colab import drive
drive.mount('/content/drive')
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
```

```

from sklearn.metrics import accuracy_score, precision_score,
recall_score, f1_score, confusion_matrix, classification_report
1. Load Dataset
df = pd.read_csv('spam.csv', encoding='latin-1')[['v1', 'v2']]
df.columns = ['label', 'message']
df['label_num'] = df['label'].map({'ham': 0, 'spam': 1})
2. Text Preprocessing & TF-IDF Vectorization
vectorizer = TfidfVectorizer(stop_words='english', max_features=3000)
X = vectorizer.fit_transform(df['message'])
y = df['label_num']
3. Train-Test Split (80/20)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
4. Model Training & Evaluation
model = MultinomialNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
5. Output Metrics
print("=== METRICS RESULTS ===")
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print(f"Precision: {precision_score(y_test, y_pred):.4f}")
print(f"Recall: {recall_score(y_test, y_pred):.4f}")
print(f"F1-Score: {f1_score(y_test, y_pred):.4f}\n")
print(classification_report(y_test, y_pred, target_names=['Ham',
'Spam']))
6. Generate Confusion Matrix Plot
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(5, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Ham',
'Spam'], yticklabels=['Ham', 'Spam'])
plt.title('Spam Classifier - Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=300)
plt.show()

```

4.Results & Visual Screenshots:

=== METRICS RESULTS ===

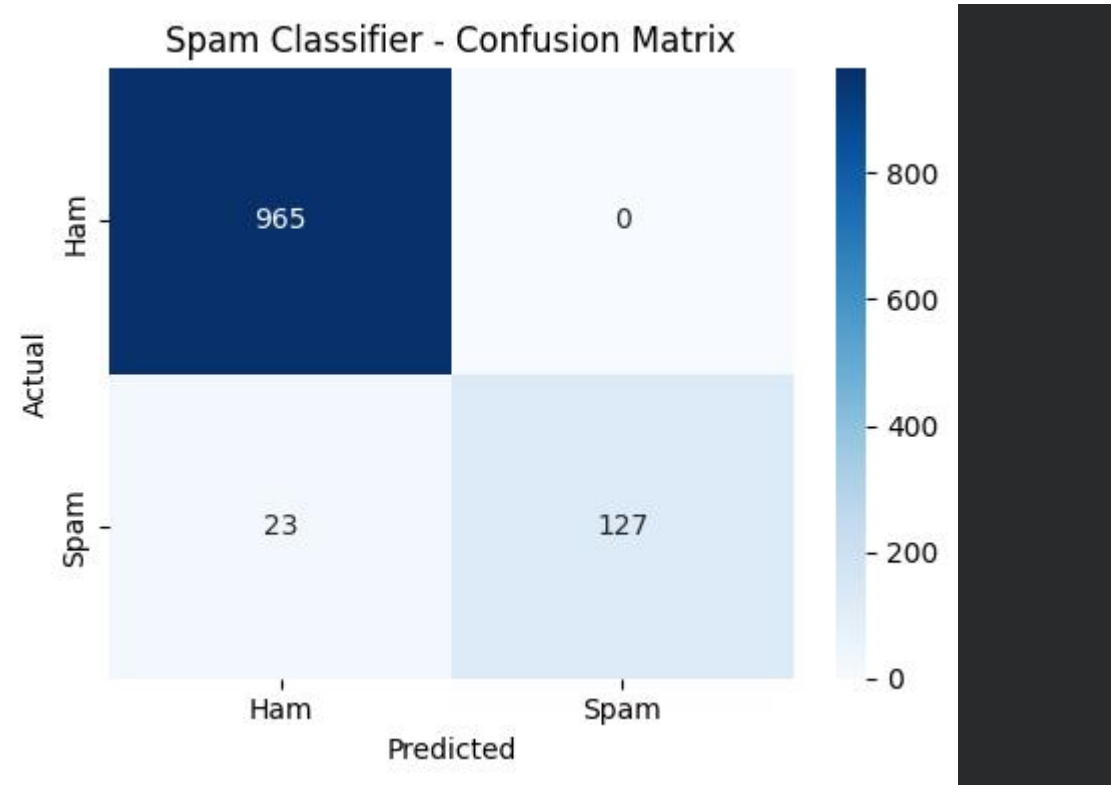
Accuracy: 0.9794

Precision: 1.0000

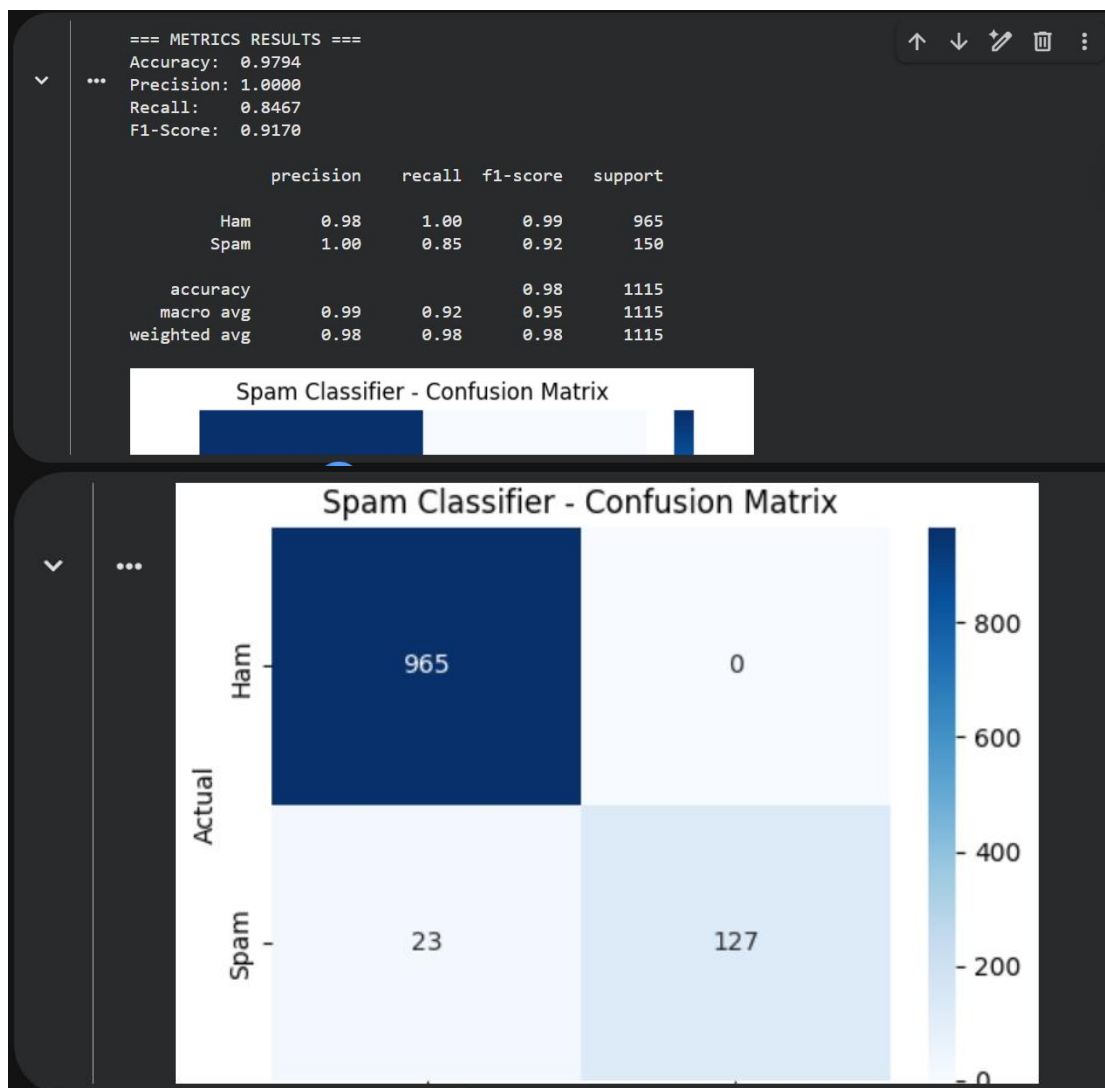
Recall: 0.8467

F1-Score: 0.9170

	precision	recall	f1-score	support
Ham	0.98	1.00	0.99	965
Spam	1.00	0.85	0.92	150
accuracy			0.98	1115
macro avg	0.99	0.92	0.95	1115
weighted avg	0.98	0.98	0.98	1115



5.Screenshots:



6.Local Testing Strategy (In Google Colab):

6.1 Code :

Function to test custom email text

```
def predict_email(text):
    text_vectorized = vectorizer.transform([text])
    prediction = model.predict(text_vectorized)[0]
    confidence = model.predict_proba(text_vectorized).max() * 100

    result = "SPAM" if prediction == 1 else "HAM (Legitimate)"
    print(f"Text: '{text}'")
    print(f"Result: {result} ({confidence:.2f}% confidence)\n")

Test Cases
predict_email("CONGRATULATIONS! You won a $1,000 Walmart gift card.
Click here to claim now!")
```

```
predict_email("Hey Kashif, let me know when you are free for the  
project discussion.")
```

6.2 Screenshot:

```
*** Text: 'CONGRATULATIONS! You won a $1,000 Walmart gift card. Click here to claim now!'  
Result: SPAM (89.52% confidence)  
  
Text: 'Hey Kashif, let me know when you are free for the project discussion.'  
Result: HAM (Legitimate) (98.40% confidence)
```